

Digital Health Record Platform — Full Workflow Guide

A Complete Non-Technical & Deep-Dive Explanation of System Architecture, Portals & Capabilities

1. System Overview & Core Purpose

The **Digital Health Record Platform** is a unified healthcare management ecosystem designed to eliminate paper record loss, fragmented medical histories, and slow report deliveries. Built with a modern **Flutter Frontend** and a high-performance **Python Django REST Backend**, the platform seamlessly connects four distinct stakeholders: **Patients, Doctors, Diagnostic Labs, and Hospitals**.

User Portal / Role	Primary Responsibility	Key Capabilities & Access Limits
Patient Portal	Personal Health Locker & Privacy Control	Full control over health records, OTP login, Emergency Card, Family member access
Doctor Portal	Clinical Care & Prescriptions	Search patient directory, view historical medical charts (if consented), prescribe
Diagnostic Lab Portal	Test Ingestion & Results	Receive pending test orders, fill verified parameter values, publish lab results
Hospital Care Portal	Inpatient & Ward Care	Manage ward beds, admit inpatients, track ICU/General stays, discharge planning

2. Detailed Patient Workflow (Step-by-Step)

When a patient opens the application, here is exactly how their journey works from start to finish:

Step 1: Simple Password-Less Authentication (Email / Mobile OTP)

- **Login & Verification:** The patient enters their email or phone number. The backend instantly generates a secure 6-digit One-Time Password (OTP) valid for 5 minutes.
- **First-Time Registration:** If it's a new user, they complete their basic profile (Full Name, Gender, Age/DOB, Blood Group, Allergies, Emergency Contact).

Step 2: Patient Dashboard & Quick Glance

- **Health Vitals Tracker:** Shows latest recorded metrics (Blood Pressure, Heart Rate, SpO2, Blood Sugar, BMI) with color-coded status chips (Normal, Elevated, High).
- **Quick Actions:** 1-tap buttons to show Emergency Card, view recent lab reports, book doctor appointments, or check active consent approvals.

Step 3: Digital Health Locker & Medical Records

- **Centralized Vault:** Every prescription written by a doctor, every lab report published by a lab, and every hospital discharge summary automatically lands here.
- **What Patient Can Do:** View records, filter by type (Prescription, Lab Report, Discharge Summary), search keywords, download PDF reports.
- **What Patient Cannot Do:** Alter or tamper with official diagnostic results published by verified labs or doctors (ensuring legal data integrity).

Step 4: Unified Medical Timeline

- **Chronological Stream:** Combines events across all clinics, diagnostic centers, and hospitals into a single chronological timeline (e.g. June 12: CBC Test at Lab Hub -> June 14: OPD Consultation at Patel Clinic -> July 01: Hospital Discharge at City Hospital).

Step 5: Patient Privacy & Consent Control Center

- **Strict Data Privacy:** Patient health records are locked by default. Doctors cannot peek into historical medical records without patient permission.
- **Approving Access:** When a doctor requests record access, the patient receives an access request. When approved, access is granted for exactly 24 hours.
- **1-Tap Instant Revocation:** At any time, the patient can hit 'Revoke Access' to instantly cut off doctor access immediately.

Step 6: Family Account Management & Emergency Card

- **Dependents Care:** Patients can add family members (children, elderly parents, spouse) and manage their health records under one master account.
- **Emergency Card:** Designed for critical situations, presenting Blood Group, Known Allergies, Emergency Contacts, and a scannable Patient QR code.

3. Detailed Doctor Workflow (Step-by-Step)

Step 1: Doctor Login & OPD Dashboard

- Doctors log in with verified medical credentials (Registration Number, Specialization, Clinic/Hospital Affiliation).
- The doctor dashboard lists today's appointments, queue status, and pending patient access approvals.

Step 2: Patient Directory Search & Registration

- **Instant Search:** Doctors can search patients by Name, Phone Number, Email, or Patient Code (e.g., PAT-4726A2).
- **On-the-Fly Registration:** If a new walk-in patient visits, the doctor can register them directly. This automatically creates the patient account and pre-authorizes the doctor for 24 hours for immediate consultation.

Step 3: Consent-Gated Clinical Chart Access

- **Unlocked Chart:** If active consent exists (or for newly registered walk-ins), the doctor sees full medical history, past prescriptions, vitals trends, and lab reports.
- **Locked Chart:** If consent is missing or expired, historical records display a security lock. The doctor clicks 'Request Access', triggering a notification to the patient.

Step 4: Writing Digital Prescriptions & Automated Synchronization

- Doctor enters Clinical Diagnosis, Notes, and prescribes medicines (Medicine Name, Dosage e.g., 1-0-1, Duration e.g., 7 Days).
- **Auto-Sync Magic:** As soon as the doctor clicks 'Save Prescription', the backend immediately generates a formal record and injects it into the patient's personal health locker in real time!

4. Detailed Diagnostic Lab Workflow (Step-by-Step)

Step 1: Order Queue Management

- Lab technicians log in to view pending diagnostic test orders placed by doctors or requested by patients.

Step 2: Verification & Publishing Results

- Lab staff input individual test parameters (e.g. Hemoglobin: 14.2 g/dL, Fasting Blood Sugar: 110 mg/dL).
- Abnormal flags (High/Low) are set automatically or manually. Once verified, clicking 'Publish Report' instantly transfers the official verified report into the patient's digital locker.

Step 3: High-Volume Batch CSV Processing

- Diagnostic centers processing hundreds of daily samples can upload a single CSV spreadsheet.
- The backend engine automatically matches each row to the patient email. If a patient is not in the database, it automatically creates their account and attaches their lab report seamlessly.

5. Detailed Hospital Care Workflow (Step-by-Step)

Step 1: Ward & Inpatient Bed Allocation

- Hospital administration tracks bed availability across General Wards, ICU, and Emergency Units.
- When admitting a patient, staff record Ward Name, Bed Number, Attending Doctor, and Initial Diagnosis.

Step 2: Patient Discharge & Summary Auto-Sync

- Upon recovery, hospital staff execute the discharge procedure and input clinical discharge summary notes.
- Hitting 'Discharge' frees up the hospital bed and automatically transmits a formatted Discharge Summary into the patient's digital locker.

6. How Everything Works Behind The Scenes (In Simple Words)

1. The User Interface (Frontend): Built with **Flutter**, delivering a fast, responsive experience across mobile phones and web browsers. It never talks directly to the raw database; it communicates via secure API requests.

2. The Application Engine (Backend): Powered by **Python Django REST Framework**. It acts as the traffic controller, verifying security tokens, checking permissions, validating inputs, and executing database queries.

3. Multi-Role Security Guard: Ensures that a Patient can only see their records, a Doctor can only see patients who granted consent, a Lab can only publish test results, and a Hospital can only manage ward admissions.

4. Automated Synchronization Engine: Eliminates manual data entry between systems. When an action occurs in one portal (e.g. Doctor writes prescription or Lab publishes test), background signals automatically update the Patient's personal locker instantly.